

LAB 3 - Routing Protocol

NAME: NUR FIRZANA BINTI BADRUS HISHAM

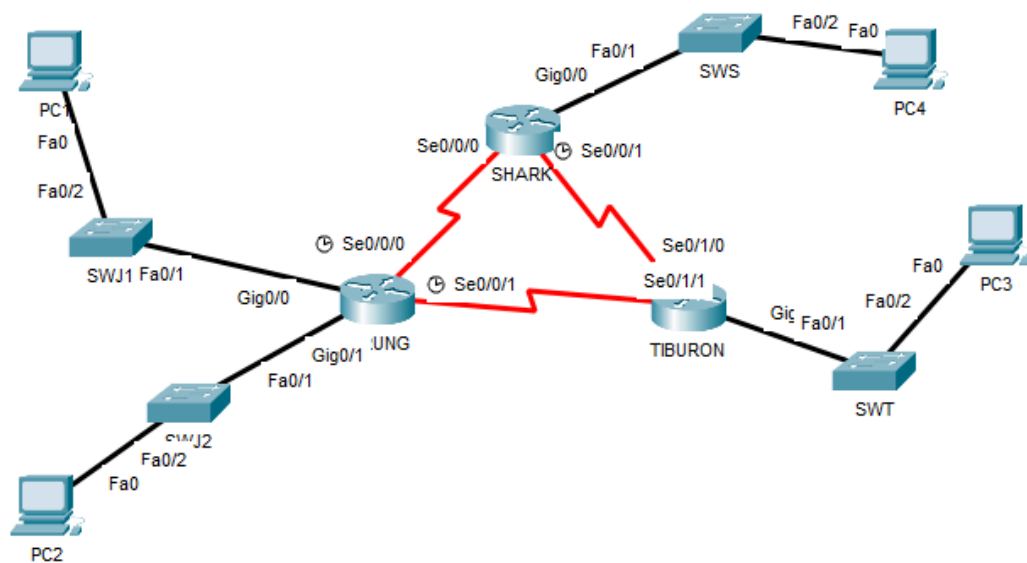
MATRIC NUM: A23CS0156

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.129	255.255.255.224	-
4		G0/1	172.18.110.161	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.158	255.255.255.224	172.18.110.129
12	PC2	-	172.18.110.190	255.255.255.224	172.18.110.161
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show **your workings here** and **complete Table 2** with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, **configure the devices** with the appropriate information.

Calculation of the IP Address and Subnet Mask:

SHARK (hosts used: 60)

$$2^x = 60 \rightarrow 2^6 = 64 \text{ (extra 4)} \quad /26$$

$$32 - 6 = 26$$

1010 1100 . 0001 0010 . 0110 1110 . 00000000

_____ . 0000 0000 [0]
_____ . 0011 1111 [63]

range: 172.18.110.0
172.18.110.63

/26

1111 1111 . 1111 1111 . 1111 1111 . 1100 0000

255.255.255.192

TIBURON (hosts used : 50)

$$2^x = 50 \rightarrow 2^6 = 64 \text{ (extra 14)}$$

$$32 - 6 = 26 \quad /26$$

1010 1100 . 0001 0010 . 0110 1110 . 00000000

subnet

host
portion

_____ . 01000000 [64]

_____ . 01111111 [127]

range: 172.18.110.64
172.18.110.127

/26

1111 1111 . 1111 1111 . 1111 1111 . 1100 0000

255.255.255.192

JERUNG LAN₂ (hosts used : 30)

$$2^x = 30 \rightarrow 2^5 = 32$$

$$32 - 5 = 27 \quad /27$$

$1010\ 1100.0001\ 0010.0110\ 1110.0000\ 00000$
 subnet host portion
 _____ . 100 0 0000 [128]
 _____ . 100 1 1111 [159]

range 172.18.110.128
172.18.110.159

127

11111111.11111111,11111111,11100000
255.255.255.224

DESIGNING LAN 1 (hosts used : 20)

$$2^x = 20 \rightarrow 2^5 = 32$$

$$32 - 5 = 27 \quad /27$$

1010 1100 . 0001 0010 . 0110 1110 . 0000 0000

subnet | host
portion

_____ . 101 0 0000 [160]

_____ . 101 1 1111 [191]

range 172.18.110.160
 172.18.110.191

/27

11111111 . 11111111 . 11111111 . 1110 0000
255.255.255.224

JERUNG-SHARK (hosts used: 2)

$$2^x = 2 \rightarrow 2^2 = 4$$

$$32 - 2 = 30 \quad / 30$$

1010 1100.0001 0010.0110 1110.00000000

_____ . 1100 0000 [192]

 , 1100 0011 [195]

range 172.18.110.192

172. 18. 110. 195

Subnet mask

130

1111 1111, 1111 1111, 1111 1111, 1111 1100

255, 255, 255, 252

JERUNG - TIBURON (hosts used: 2)

$$2^x = 2 \rightarrow 2^2 = 4$$

$$32 - 2 = 30 \quad /30$$

1010 1100 . 0001 0010 . 0110 1110 . 00000000

subnet host portion

_____ . 1100 0100 [196]

_____ . 1100 0111 [199]

range : 172.18.110.196
172.18.110.199

Subnet mask

/30

1111 1111 . 1111 1111 . 1111 1111 . 1111 1100

255.255.255.252

SHARK-TIBURON (hosts used: 2)

$$2^x = 2 \rightarrow 2^2 = 4$$

$$32 - 2 = 30 \quad /30$$

1010 1100 . 0001 0010 . 0110 1110 . 00000000

subnet

host
portion

_____ . 1100 10 00 [200]

_____ . 1100 10 11 [203]

range 172.18.110.200
 172.18.110.203

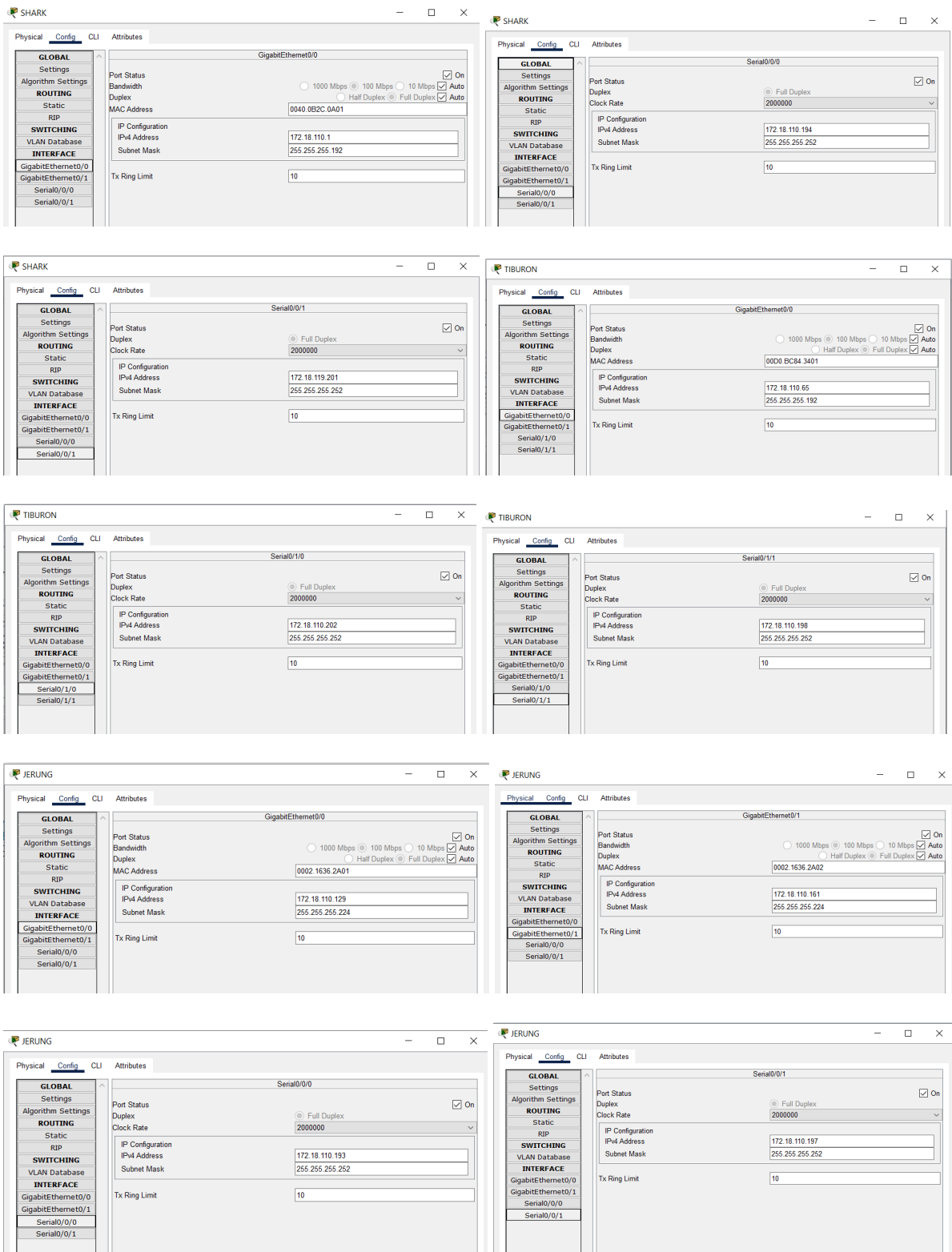
subnet mask

/30

1111 1111 . 1111 1111 . 1111 1111 . 1111 1100

255.255.255.252

Screenshot of the work progress:



Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

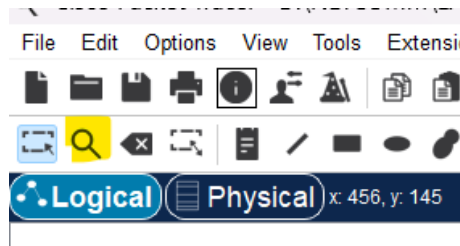


Figure B

A screenshot of a window titled 'Routing Table for SHARK'. It displays a table with the following data:

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Here is the routing table for each router:

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.119.200/30 is directly connected, Serial0/0/1
L       172.18.119.201/32 is directly connected, Serial0/0/1
```

shark router (CLI)

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.119.200/30	Serial0/0/1	---	0/0
L	172.18.119.201/32	Serial0/0/1	---	0/0

shark (PT Tool)

```
TIBURON>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

Tiburon Router (CLI)

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Tiburon Router (PT Tool)

```

JERUNG>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C    172.18.110.128/27 is directly connected, GigabitEthernet0/0
L    172.18.110.129/32 is directly connected, GigabitEthernet0/0
C    172.18.110.160/27 is directly connected, GigabitEthernet0/1
L    172.18.110.161/32 is directly connected, GigabitEthernet0/1
C    172.18.110.192/30 is directly connected, Serial0/0/0
L    172.18.110.193/32 is directly connected, Serial0/0/0
C    172.18.110.196/30 is directly connected, Serial0/0/1
L    172.18.110.197/32 is directly connected, Serial0/0/1
  
```

Jerung Router (CLI)

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Jerung Router (PT Tool)

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=16ms TTL=127
Reply from 172.18.110.190: bytes=32 time<1ms TTL=127
Reply from 172.18.110.190: bytes=32 time=11ms TTL=127

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 9ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

- Ping PC 2 from PC1

The ping was partially successful, but with 25% packet loss. This could be due to the network congestion which is the network might be overloaded, and some packets may be dropped.

- Ping PC 4 from PC1

The router is unable to find a route to reach the destination because the result shows Destination host unreachable. The routing table on the router is incomplete or missing a route to the subnet that contains PC4.

5. What needs to be done to ensure all PCs can ping each other successfully?

Ensure all the PCs have the correct IP Address and subnet mask and also ensure each PC's default gateway is correctly set to the IP address of the router interface connected to its subnet. Ensure the RIP version 2 is properly configured on all routers and all relevant networks are included in RIP configuration.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```
JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#
```

Figure E

- a. What can you say about the addresses used in the 'network' instructions in Figure E?

The addresses used in the network commands on Jerung are network addresses representing the subnets connected to the router. In RIP configuration, network addresses are used for each subnet to connect to the router.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/30	GigabitEthernet0/0	---	0/0	R	172.18.110.64/30	Serial0/0/1	172.18.110.198	120/1
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1	C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0	C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0	L	172.18.110.197/32	Serial0/0/1	---	0/0
					R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Routing Tables for Tiburon and Jerung

- a. What are the changes seen in TIBURON?

After configuring the RIP on both routers Jerung and Tiburon, the routing table in Tiburon displays newly learned networks from Jerung, marked with the R for RIP. These entries represent the subnets learned via RIP protocol.

- b. What are the Networks with type R in TIBURON and JERUNG?

- TIBURON
 - 172.18.110.128/27
 - 172.18.110.160/27
 - 172.18.110.192/30
- JERUNG
 - 172.18.110.64/30
 - 172.18.110.200/30

- c. Ping PC3 from PC1. Was it successful?

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=11ms TTL=126
Reply from 172.18.110.126: bytes=32 time=18ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 18ms, Average = 10ms

```

The ping was successful with 25% packet loss.

- d. Ping PC4 from PC1. Was it successful?

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time=12ms TTL=126
Reply from 172.18.110.62: bytes=32 time=11ms TTL=126
Reply from 172.18.110.62: bytes=32 time=11ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 12ms, Average = 11ms

```

The ping was successful with 25% packet loss.

- e. Explain the reasons for your answer.

RIP version 2 is configured correctly on all routers. Each router knows how to reach every subnet in the network, enabling successful communication between PCs on different subnets.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```

SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.1
SHARK(config-router)#network 172.18.110.193
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#

```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
R	172.18.110.64/30	Serial0/0/1	172.18.110.202	120/1	
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Reply from 172.18.110.158: bytes=32 time=8ms TTL=126
Reply from 172.18.110.158: bytes=32 time=12ms TTL=126
Reply from 172.18.110.158: bytes=32 time=14ms TTL=126
Reply from 172.18.110.158: bytes=32 time=16ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 16ms, Average = 12ms
```

PC1

```
Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=40ms TTL=126
Reply from 172.18.110.190: bytes=32 time=41ms TTL=126
Reply from 172.18.110.190: bytes=32 time=12ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 41ms, Average = 23ms
```

PC2

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=50ms TTL=125
Reply from 172.18.110.126: bytes=32 time=12ms TTL=125
Reply from 172.18.110.126: bytes=32 time=24ms TTL=125
Reply from 172.18.110.126: bytes=32 time=14ms TTL=125

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 50ms, Average = 25ms
```

PC3

PC	Ping result
1	Success
2	Success
3	Success

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)

11000000.10101000.00000001.00000001

^^

so, subnet mask is 255.255.255.0

- b. As this change happens, PC3 must also have a different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping PC1 from PC3

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping PC4 from PC3

No, PC4 and PC1 cannot be able to ping PC3 immediately after change because the new subnet is not yet included in the routing updates of SHARK and JERUNG routers. So, they don't have a route to reach the new subnet.

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/30	Serial0/0/1	172.18.110.202	120/1	R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	L	172.18.110.202/32	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.

The routing tables of SHARK and JERUNG have entries for the new subnet 192.168.1.0/24 because RIP was not updated to advertise the subnet.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Need to update the RIP configuration on TIBURON to advertise the new network 192.168.1.0/24.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#exit
TIBURON(config)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.158: bytes=32 time=4ms TTL=126
Reply from 172.18.110.158: bytes=32 time=11ms TTL=126
Reply from 172.18.110.158: bytes=32 time=17ms TTL=126

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 17ms, Average = 10ms
```

ping PC1 from PC3

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=14ms TTL=126
Reply from 172.18.110.190: bytes=32 time=11ms TTL=126
Reply from 172.18.110.190: bytes=32 time=10ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 14ms, Average = 11ms
```

ping PC2 from PC3

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time=12ms TTL=126
Reply from 172.18.110.62: bytes=32 time=2ms TTL=126
Reply from 172.18.110.62: bytes=32 time=11ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 8ms
```

ping PC4 from PC3

PC	Ping result
1	Success
2	Success
4	Success

REFLECTION

What have you learned in this task?

From these tasks, I have learned about the routing protocol that can change the subnet corresponding to IP address, subnet mask and gateway address. I also learned when subnet changes occur, routing protocols must be updated to propagate the new network information. Lastly, I also learned about network connectivity where a network relies on correct IP addressing, subnetting and routing configurations.